

T. A. MARRYSHOW COMMUNITY COLLEGE

DEPARTMENT OF INFORMATION TECHNOLOGY

ASSIGNMENT NAME: ONLINE PHISHING TRAINING & ANALYSIS

COURSE NAME AND NUMBER: DATA SECURITY CONCEPTS CIT245

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PRE-LAB QUESTIONS

1. **Have you ever received a message you suspected was phishing? What made you suspicious?**

Yes. I got a “ChatGPT Plus subscription” email after I already had it. The timing felt off, so I checked my account directly instead of clicking anything.

2. **Why do you think phishing continues to be one of the most successful attack methods?**

It targets human emotions such as urgency, fear, and trust, rather than just technology. People often act before thinking, especially under pressure.

3. **Which group do you think is more at risk: regular users, managers, or IT staff? Explain your answer.**

Managers. They have access to sensitive data and financial authority, and they’re often targeted with personalized, high-pressure spear-phishing emails that mimic real leadership requests.

ACTIVITY 1 - SPOT THE RED FLAGS ANALYSIS

Chosen Email: Fake PayPal receipt for unauthorized payment to "Amazing Stuff"

Email Header

• Your Receipt for Payment to Amazing Stuff 3 Yahoo/Spam ★


service@paypal.com <noreplyexmhwe9hhzdglne9ml1xqzypnedg51ny7yleqejbwe@sultanbogor.com> 2
 To: 0008812034377940108id@mail.info.cnsmr.sg3.yahoo.net 1 Sun, Jul 11 at 3:29 PM ★

▲ For your security we disabled all images and links in this email. If you believe it is safe to use, mark this message as not spam.
[Show images](#)

Email Body 1



Sun, July 11, 2021 9:29 PM

Transaction ID: 74221905U

Hello Customer,

You sent a payment of \$120.00 USD to Amazing Stuff
(helpcenter@)

It may take a few moments for this transaction to appear in your account.

Merchant Amazing Stuff helpcenter@ .com 899-552- 	Instructions to merchant You haven't entered any instructions.
Email address - confirmed ppmxtrolling @ .com	Shipping details The seller hasn't provided any shipping details yet.

Description	Unit price	Qty	Amount
Amazing Stuff Gift Card (EMAIL DELIVERY)			
Item# 1123685	\$30.00 USD	4	\$120.00 USD
Subtotal			\$120.00 USD
Total			\$120.00 USD
Payment			\$120.00 USD

Email Body 2

Charge will appear on your credit card statement as "PAYPAL "WORLDGIFT"

Payment sent to helpcenter@[REDACTED].com

Funding Sources Used (Total)

Credit Card

\$120.00 USD

If you didn't make this order or if you believe an unauthorized person is attempting to access your account, please click the button below for cancel your order.

Cancel the order

Issues with this transaction?

You have 180 days from the date of the transaction to open a dispute in the Resolution Center.

Questions? Go to the Help Center at www.paypal.com/help.

Please do not reply to this email. This mailbox is not monitored and you will not receive a response. For assistance, log in to your PayPal account and click **Help** in the top right corner of any PayPal page or please contact us toll free at 1-888-221-1161.

You can receive plain text emails instead of HTML emails. To change your Notifications preferences, log in to your account, go to your Profile, and click **My settings**.

Copyright © 1999-2021 PayPal, Inc. All rights reserved. PayPal is located at 2211 N. First St., San Jose, CA 95131.
PayPal PPX001066:1.1:c9779baffc706

Red Flag 1: Suspicious Sender Address/Domain Mismatch

- **What you see:** The sender shows as service@paypal.com but the actual email address is noreplyexmhwe9hhzdglne9ml1xqzypnedg51ny7yleqejbwe@sultanbogor.com
- **Why it is suspicious or dangerous:** This is classic email spoofing. The display name impersonates PayPal's legitimate service email, but the actual sending domain (sultanbogor.com) has no affiliation with PayPal. This mismatch indicates the email is fraudulent.
- **What a safer, legitimate version might look like:** A genuine PayPal email would come from a verified @paypal.com domain or a confirmed subdomain like @email.paypal.com, with consistent display and actual sender addresses.

Red Flag 2: Unusual/Incorrect Recipient Email Address

- **What you see:** The email is sent to 0008812034377940108id@mail.info.cnsmr.sg3.yahoo.net
- **Why it is suspicious or dangerous:** This is not a standard email format. The numeric ID format suggests this is a temporary or forwarded address, not the user's actual Yahoo email. Legitimate services send receipts to the email address on file for the account.
- **What a safer, legitimate version might look like:** A legitimate receipt would be sent to the exact email address associated with the PayPal account (e.g., yourname@gmail.com).

Red Flag 3: Generic, Impersonal Greeting

- **What you see:** "Hello Customer,"
- **Why it is suspicious or dangerous:** PayPal (and most legitimate financial services) uses personalized greetings with the account holder's name. "Hello Customer" is impersonal and suggests the sender doesn't actually know who you are, which is common in mass phishing campaigns.
- **What a safer, legitimate version might look like:** "Hello [Your First Name]," or "Dear [Your Full Name],"

Red Flag 4: Suspicious "Cancel" Button with URL Shortener

- **What you see:** The "Cancel the order" button links to a shortened URL (not visible in the text but mentioned in the analysis that it uses a URL shortening service).
- **Why it is suspicious or dangerous:** URL shorteners hide the actual destination, making it impossible to verify legitimacy before clicking. Legitimate financial institutions use direct, transparent links to their official domains. This is designed to conceal a malicious website.
- **What a safer, legitimate version might look like:** A legitimate PayPal email might direct users to log in directly at paypal.com (by typing it) and navigate to the Resolution Center, or provide a direct, verifiable link like <https://www.paypal.com/resolutioncenter/transaction/74221905U>

Red Flag 5: Incomplete/Missing Merchant Information

- **What you see:** The merchant information shows incomplete details: "helpcenter@" and "899-552-" with cut-off text.
- **Why it is suspicious or dangerous:** Legitimate PayPal receipts contain complete, verifiable merchant contact information. Incomplete details or placeholder text suggest the email was hastily crafted or is using a template with errors.
- **What a safer, legitimate version might look like:** Complete merchant name, verified email address (e.g., support@amazingstuff.com), and full customer service phone number.

Red Flag 6: Suspicious Transaction Details (Psychological Trigger)

- **What you see:** The email claims a \$120.00 payment for "Amazing Stuff Gift Card" that the recipient supposedly made.
- **Why it is suspicious or dangerous:** This creates immediate anxiety about an unauthorized charge, prompting hasty action without verification. The vague merchant name ("Amazing Stuff") and unusual item (multiple gift cards) are designed to trigger panic and override logical thinking.
- **What a safer, legitimate version might look like:** A legitimate unexpected transaction alert would provide specific merchant details (with website), clear transaction time/date, and would not pressure immediate action through a single button.

ACTIVITY 2 - REWRITE THE PHISHING EMAIL AS A LEGITIMATE MESSAGE

Subject: Receipt for Your PayPal Payment to Amazing Stuff

From: PayPal Receipts receipts@paypal.com

Dear [Customer's First Name],

Thank you for your recent PayPal payment of \$120.00 USD to Amazing Stuff. A receipt for this transaction (ID: 74221905U) has been added to your PayPal account.

To review this transaction or manage your account activity, please log in to your PayPal account directly by typing www.paypal.com into your web browser's address bar. Once signed in, you can view all recent transactions in your Activity section. If you do not recognize this payment, you can open a dispute through the Resolution Center within your account.

Sincerely,

The PayPal Team

ACTIVITY 3 - DESIGN A (SAFE) SAMPLE PHISHING EMAIL FOR TRAINING

Subject: Immediate Action Required: Your Microsoft 365 Account Will Be Locked

From: Microsoft Security Team security@microsoft-365-alerts.info

Dear User,

We have detected multiple failed login attempts to your Microsoft 365 account from an unrecognized device in a different geographical location. To prevent unauthorized access, your account has been temporarily restricted.

You must verify your identity within the next 2 hours to prevent permanent account suspension and potential data loss. This is a critical security measure.

Please click here to verify your account now: <https://secure-login.verify-microsoft365.com/auth>

After clicking the link, you will need to enter your current Microsoft 365 password and answer your security questions to regain access. Failure to complete this verification will result in all your emails and OneDrive files being permanently inaccessible.

If you did not attempt to access your account recently, this verification is especially important to secure your account from compromise.

For assistance, you can reply to this email with your account details.

Sincerely,

Microsoft Account Security Team

ACTIVITY 4 - CREATE A SIMPLE PHISHING AWARENESS TIP

How to Spot Fake Emails:

1. **Check the email address carefully** - Look at the part after the @ symbol. It should match the company's real website address.
Example: A real Chase email comes from @chase.com, NOT @chasebank.com
2. **Watch for fake sender names** - Anyone can put any name in an email. Don't trust the name alone, always check the actual email address.
3. **Be careful with links in emails** - If you get a link in an email, don't click it right away. Contact the person who sent it first using a new email or phone call to make sure it's real.

How to Spot Fake Websites:

1. **Look at the website address** - The main part before .com should be clear and simple, not a mix of random letters and numbers.
Example: chase.com is safe, but chasecom.ru is dangerous
2. **Check for the padlock icon** - Look for the lock symbol in your browser but know that bad guys can sometimes fake this too.
3. **Never ignore browser warnings** - If your browser says, "Your connection is not Private," stop immediately and don't continue to the website.

Important Safety Rules:

- **Don't click links in unexpected emails** - Even if the email looks like it's from someone you know
- **Verify first, click later** - Always check with the sender before clicking any link
- **Use safe browsers** - Use Chrome or Firefox, not Internet Explorer
- **Watch for shortened URLs** - Short links (like bit.ly) can hide dangerous websites

What to Do If You're Unsure:

If an email looks suspicious or you're not sure about a link:

1. **Don't click anything**
2. **Don't reply to the email**
3. **Report it immediately** - Forward the email to your IT department or security team
4. **Contact IT directly** - Call your help desk for help

Remember: It's always better to check twice than to become a victim of phishing. Your caution helps protect everyone in your organization.

POST-LAB REFLECTION

The easiest red flag to spot is poor spelling and grammar, as it immediately looks unprofessional. The hardest is sophisticated domain spoofing (for example, rmicrosoft.com vs microsoft.com), especially when paired with convincing branding and secure-looking HTTPS padlock icons.

To reduce phishing success, my workplace should provide short, monthly security reminders instead of yearly training, add a simple "Report Phish" button in email, and run regular simulated phishing tests with helpful feedback, without penalty. Making it safe and easy to report suspicious emails builds a stronger human defence system.

CONCLUSION

This lab strengthened my ability to identify phishing attacks by analysing red flags, rewriting malicious emails into safe communications, and creating user-friendly security guidance. I learned that phishing succeeds by exploiting human emotions like fear and urgency, not just technical weaknesses. The most effective defence combines technical controls with ongoing user education, fostering a culture where everyone knows how to spot, avoid, and report suspicious messages.